Def. A continuous real-valued $f: D \to \mathbb{R}$ on a domain $D \subseteq \mathbb{C}$ is called harmonic if:
f is partially twice continuously differentiable ($f \in C^2$) satisfying

$$f_{xx} + f_{yy} = 0. \quad \text{(It is called the Laplace's equation.)}$$

Thm. Let $f = u + iv$ be an holomorphic on a domain D.
Then, both u and v are harmonic on D.

Proof. By the Cauchy-Riemann equation, $u_x = v_y$ and $u_y = -v_x$, so that

$$f' = u_x + i v_x = u_x - i \cdot u_y = v_y + i \cdot v_x$$

$$\hookrightarrow f'' = u_{xx} - i \cdot u_{yx} = v_{yx} + i v_{xx}$$
$$= u_{xx} - i \cdot u_{xy} = v_{xy} + i v_{xx} \quad \Leftarrow \text{Fubini's Theorem}$$
$$= u_{xx} - i \cdot v_{yy} = -u_{yy} + i v_{xx}$$

Therefore, $u_{xx} = -u_{yy}$ and $-v_{yy} = v_{xx}$.

Thm. Suppose that f is analytic on D satisfying $|f(z)|$ is constant on D.
Then f is constant in D.

Proof. Denote $f = u + iv$. By the assumption,

$|u + iv| = C$ for all $z \in D$, so $u^2 + v^2 = C^2$.

Differentiating with respect to real and imaginary, then

$$2uu_x + 2vv_x = 2uu_y + 2vv_y = 0,$$

substituting the terms in Cauchy-Riemann equation, $u_x = v_y$ and $v_x = -u_y$.

$uu_x - vu_y = uu_y + vu_x = 0$, multiplying u to the left side, then

$uuu_x - uvu_y = uuu_x + vvu_x = u_x(u^2 + v^2) = 0$, so $u_x = 0$.
(If $u_x \neq 0$, then $u^2 + v^2 = C^2 = 0$, it means f is identically zero).
Similarly, $u_y = v_x = v_y = 0$, so f' is zero, it gives the f is constant. $\boxtimes$

Another proof of: $|f|$ is constant $\Rightarrow f$ is constant.

Lemma. If f is real-valued and holomorphic on a domain D, then f is constant.

Proof. Given $z_0 \in D$,

$$\lim_{h \to 0} \frac{f(z_0+h) - f(z_0)}{h}$$ exists by the hypothesis, so

letting $h \to 0$ on pure real axe, then the derivative is also pure real, since the f can have real value. Similarly, letting $h \to 0$ on pure imaginary, the derivative must be pure imaginary. So, $f'(z_0)$ must be zero.

Since z_0 was arbitrary, $f' \equiv 0$ on D, so f is constant. ▨

Now, suppose that $|f| \equiv C$ for some $C \in \mathbb{C}$.

If $C = 0$, then $f = 0$, so there is nothing to prove.

If $C > 0$, then $|f|^2 = f\bar{f} = C^2$, so $\bar{f} = \frac{C^2}{f}$ is analytic on D.

Therefore, $\operatorname{Re} f = \frac{f+\bar{f}}{2}$ and $\operatorname{Im} f = \frac{f-\bar{f}}{2i}$

both are holomorphic, so constant, by the Lemma above. ▨

And, using the Lemma, show that a fun result:

Thm. If image of f lies on a line in $\mathbb{C}$ on a domain D, then f is constant.

Proof. Let

「Thm.

If $u: D \to \mathbb{R}$ is harmonic on a simply connected domain D,
then there exists an analytic function f on D satisfying $\mathrm{Re}(f) = u$ on D.

Proof. Define $g(z) = u_x(z) - i \cdot u_y(z)$. Then, since the u is harmonic and so $u \in C^2$,

$$u_{xx} = -u_{yy}, \quad \text{and} \quad u_{xy} = u_{yx} \text{ by the Fubini's Theorem} \quad \text{---} (*)$$

Actually, the $(*)$ is a Cauchy-Riemann equation for g, so g is analytic on D,
Since $u \in C^2$. Let $z_0 \in D$ be fixed. Define

$$F(z) = \int_{z_0}^{z} g(\mu)\, d\mu \quad \text{on } D.$$

So, F is defined as integral of an analytic function, so F is analytic. And,

$$F'(z) = g(z) = u_x(z) - i \cdot u_y(z)$$

$$= \frac{\partial}{\partial x} \mathrm{Re}\, F(z) - i \cdot \frac{\partial}{\partial y} \mathrm{Re}\, F(z)$$

So that $\mathrm{Re}\, F(z) = u(z) + C$ for some constant $C \in \mathbb{R}$, therefore

$$f(z) = F(z) - C = \int_{z_0}^{z} u_x(\mu) - i \cdot u_y(\mu)\, d\mu - C$$

is we desired. ☒

「Example. Let $u(x,y) = \sin x \cdot \cosh y + \cos x \cdot \sinh y + x^2 - y^2 + 2xy$.
By the theorem, there is analytic $f = u + iv$, so

$$f'(z) = u_x - i u_y = \cos x \cdot \cosh y - \sin x \sinh y + 2x + 2y$$
$$- i \cdot (\sin x \cdot \sinh y + \cos x \cdot \cosh y - 2y + 2x)$$

By definition, $\cos(iy) = \dfrac{e^{i^2 y} + e^{-i^2 y}}{2} = \cosh(y)$ and $\sin(iy) \cdot (-i) = \sinh y$,

So we can simplify

$$f'(z) = \cos x \cdot \cos(iy) + i \cdot \sin x \cdot \sin(iy) + 2x - 2y$$
$$- i \cdot (-i \cdot \sin x \cdot \sin(iy) + \cos x \cdot \cos(iy) - 2y + 2x)$$
$$= (1 - i)(\cos x \cdot \cos(iy) - \sin x \cdot \sin(iy) + 2z)$$
$$= (1 - i) \cdot (\cos z + 2z)$$

So, $f(z) = (1 - i) \cdot (\sin z + z^2) + C$.

「Thm.

A harmonic and bounded function must be a constant.

Proof. Suppose that $u(z)$ is harmonic and bounded.

By the theorem, there is an analytic $f: \mathbb{C} \to \mathbb{C}$ such that $\mathrm{Re}\, f = u$,

Then, $g(z) = e^{f(z)}$ satisfies $|g(z)| = e^{u(z)}$

is bounded, since u is bounded. Therefore, by the Liouville's theorem, the g is constant, also f, so also u. $\boxtimes$